Hackathon Challenge

Apply the Full ML Pipeline to a New Clinical Dataset

QLS–MiCM Workshop

The Goal

Build the **best-performing classifier** for predicting *diabetes risk* on a new clinical dataset.

You will:

- use everything learned so far.
- explore models *creatively*.
- compare performance fairly.
- explain your reasoning clearly.

What You're Given

- A dataset: **clinical risk**.
- A **baseline pipeline** with preprocessing + logistic regression.
- A train/test split.
- Code that computes performance metrics.

This is your starting point.

What You Can Change (Important)

You may modify **only the model step** inside the existing pipeline.

You may:

- swap in any scikit-learn classifier.
- adjust its hyperparameters.
- run multiple experiments.
- document your choices.

You may **NOT**:

- change preprocessing.
- change the train/test split.
- alter or rebalance the data.
- engineer new features.
- manually inspect or label data.
- use external libraries or pretrained models.

All teams compete under the **same constraints**.

How We Will Evaluate

Each team will present:

- **Model used**.
- **Final test-set score (ROC AUC)**.
- **Short explanation** of what you tried and why.

Ranking is based on:

1. **ROC AUC**.
2. Clarity of reasoning.
3. Efficient exploration (not random guessing).

Team Workflow

1. Record what you tried and why.
2. Keep your notes short and clear.
3. When time is up -> report your best score.

Good Luck!.

May the best model win.